



# Caringbah High School

Year 12 2022

## Mathematics Extension 2

HSC Course

### Assessment Task 4 – Trial HSC Examination

#### General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black or blue pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 100

#### **Section I** 10 marks

Attempt Questions 1-10

Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

#### **Section II** 90 marks

Attempt Questions 11-16

Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

Name: \_\_\_\_\_

Class: \_\_\_\_\_

| Marker's Use Only |            |     |     |     |     |     |       |
|-------------------|------------|-----|-----|-----|-----|-----|-------|
| Section I         | Section II |     |     |     |     |     | Total |
| Q 1-10            | Q11        | Q12 | Q13 | Q14 | Q15 | Q16 |       |
| /10               | /15        | /15 | /15 | /15 | /15 | /15 | /100  |

## Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet (page 13) for Questions 1–10

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1. Let the point  $P$  on an Argand diagram represent the complex number  $z$ . After being multiplied by another complex number,  $\omega$ ,  $P$  is rotated  $90^\circ$  anti-clockwise and  $z$  is enlarged by a factor of 3. Which of the following is the value of  $\omega$ ?
- (A)  $3i$   
(B)  $-3i$   
(C)  $e^{-3i}$   
(D)  $3e^{\frac{\pi}{2}i}$
2. A particle moves in a straight line so that its displacement,  $x$  metres, at any time,  $t$  seconds, is given by  $x = 5 \sin 3t + 12 \cos 3t$ . What is the speed of the particle as it passes through the centre of its motion?
- (A) 12 m/s  
(B) 13 m/s  
(C) 39 m/s  
(D) 117 m/s
3. Which of the following is equivalent to the expression  $\frac{12x-3}{(x-2)(x^2-3x+2)}$ ?
- (A)  $\frac{9}{x-1} + \frac{9}{x-2} - \frac{21}{(x-2)^2}$   
(B)  $\frac{9}{x-1} + \frac{18}{x-2} - \frac{21}{(x-2)^2}$   
(C)  $\frac{9}{x-1} - \frac{9}{x-2} + \frac{21}{(x-2)^2}$   
(D)  $\frac{9}{x-1} - \frac{18}{x-2} + \frac{21}{(x-2)^2}$

4. Which of the following is the vector equation of a line that passes through the

point  $(1, 3, -2)$  and is perpendicular to the line  $\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$ ?

(A)  $\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$

(B)  $\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$

(C)  $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}$

(D)  $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ -3 \end{pmatrix}$

5. The acceleration of a particle is given by  $a = e^{-3t}$  m/s<sup>2</sup> where the particle has an initial velocity of 2 m/s. What is the terminal velocity of the particle?

(A)  $1.\dot{3}$  m/s

(B)  $1.\dot{6}$  m/s

(C)  $2.\dot{3}$  m/s

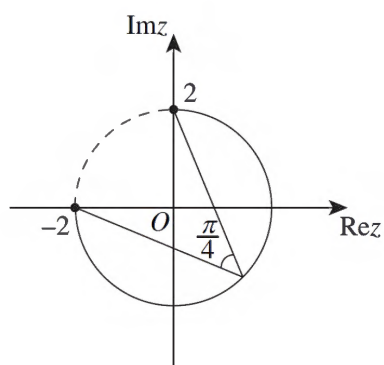
(D)  $2.\dot{6}$  m/s

**Section I continues on page 5**

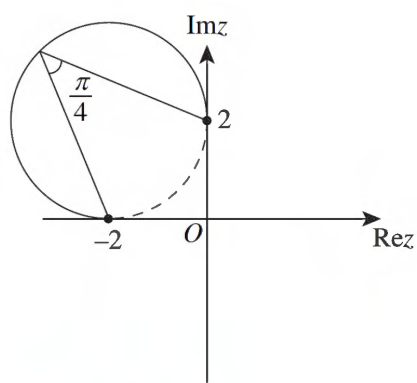
6. Which of the following diagrams best represents the solutions to the equation

$$\arg\left(\frac{z-2}{z+2i}\right) = \frac{\pi}{4}?$$

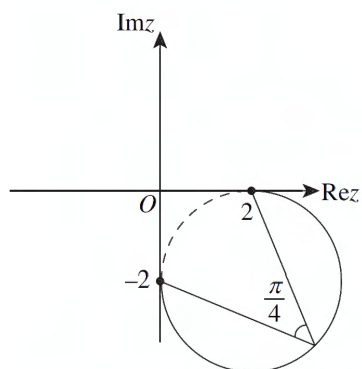
(A)



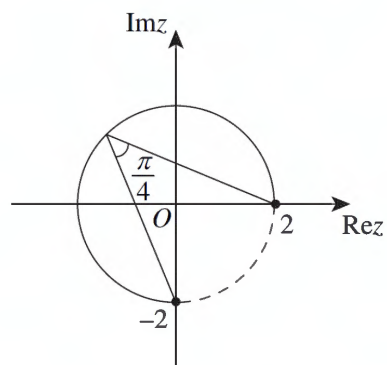
(B)



(C)



(D)



7. The points  $A$ ,  $B$  and  $C$  are collinear where  $\overrightarrow{OA} = \underline{i} - \underline{j}$ ,  $\overrightarrow{OB} = -3\underline{j} - \underline{k}$  and  $\overrightarrow{OC} = 2\underline{i} + a\underline{j} + b\underline{k}$  for some constants  $a$  and  $b$ . What are the values of  $a$  and  $b$ ?
- (A)  $a = -1$  and  $b = -1$   
 (B)  $a = -1$  and  $b = 1$   
 (C)  $a = 1$  and  $b = -1$   
 (D)  $a = 1$  and  $b = 1$
8. Consider the statement:  $\exists x \in \mathbb{R}, \ln x = 1 \text{ and } x > 2$   
 Which of the following is the negation of the statement?
- (A)  $\exists x \in \mathbb{R}, \ln x \neq 1 \text{ or } x \leq 2$   
 (B)  $\exists x \in \mathbb{R}, \ln x \neq 1 \text{ and } x \leq 2$   
 (C)  $\forall x \in \mathbb{R}, \ln x \neq 1 \text{ or } x \leq 2$   
 (D)  $\forall x \in \mathbb{R}, \ln x \neq 1 \text{ and } x \leq 2$
9. If  $\int_1^4 f(x) dx = k$  for some constant  $k$ , what is the value of  $\int_1^4 f(5-x) dx$ ?
- (A)  $-k$   
 (B)  $5-k$   
 (C)  $k+5$   
 (D)  $k$
10. Which of the following best describes the path of a particle with the following parametric equations?
- $$\begin{cases} x = t \sin t \\ y = t \cos t \\ z = t \end{cases}$$
- (A) spiral around the  $z$ -axis, traversing in a clockwise direction  
 (B) spiral around the  $z$ -axis, traversing in an anticlockwise direction  
 (C) helix around the  $z$ -axis, traversing in a clockwise direction  
 (D) helix around the  $z$ -axis, traversing in an anticlockwise direction

**End of Section I**

## Section II

60 marks

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the complex numbers  $z = 2 + 2\sqrt{3}i$  and  $w = 2\sqrt{3} - 2i$ .
- (i) Find  $z$  and  $w$  in modulus-argument form. 2
  - (ii) Find  $zw$  and  $\frac{z}{w}$  in modulus-argument form. 2
  - (iii) Describe the relationship between  $z$  and  $w$  geometrically. 1
- (b) Consider the equation  $f(z) = 0$  where  $f(z) = z^3 - 11z^2 + 55z - 125$ .
- (i) Find the three roots of the equation in the form  $a + ib$ , where  $a$  and  $b$  are real. 3
  - (ii) Show that the points  $A$ ,  $B$  and  $C$  in the Argand diagram representing these roots lie on a circle of the form  $|z| = k$  for some constant  $k$ , and find the area of  $\triangle ABC$ . 2
- (c) (i) Find constants  $A$  and  $B$  such that 2
- $$A(3 \sin x + 2 \cos x) + B(3 \cos x - 2 \sin x) = 8 \sin x + 14 \cos x$$
- (ii) Hence, find the exact value of  $\int_0^{\frac{\pi}{2}} \frac{8 \sin x + 14 \cos x}{3 \sin x + 2 \cos x} dx$  3

**End of Question 11**

**Question 12** (15 marks) Use a SEPARATE writing booklet.

- (a) We are given that  $\frac{1}{a}$ ,  $\frac{1}{b}$  and  $\frac{1}{c}$  are three consecutive terms in a geometric series, where  $a, b, c \in \mathbb{R}$ . Show that  $a^2 + c^2 \geq ab + bc$ . 3
- (b) A body of unit mass falls under gravity through a resisting medium. The body falls from rest from a height above the ground. The resistance to its motion is  $\frac{1}{100}v^2$  where  $v$  m/s is the speed of the body when it has fallen a distance  $x$  m. The acceleration due to gravity is  $g$  m/s<sup>2</sup>.
- (i) Show that the equation of motion of the body is  $\ddot{x} = g - \frac{1}{100}v^2$  1
- (ii) Show that the terminal velocity  $V$  m/s of the body is given by  $V = 10\sqrt{g}$  1
- (iii) Hence show that  $v^2 = V^2 \left(1 - e^{-\frac{x}{50}}\right)$  3
- (iv) Find the distance fallen in metres until the body reaches a velocity equal to half of its terminal velocity. 2  
(You may assume this occurs before the body reaches the ground.)
- (c) (i) Use the substitution  $u = \frac{1}{x}$  to show that  $\int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx = 0$  2  
for any constant  $a > 0$ .
- (ii) Hence, find  $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\tan^{-1} x}{x} dx$  in simplest exact form. 3

**End of Question 12**

**Question 13** (15 marks) Use a SEPARATE writing booklet.

- (a) Consider the lines  $L_1$  and  $L_2$ , determined by the vector equations

$$L_1: \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad L_2: \mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Show that  $L_1$  and  $L_2$  intersect and are perpendicular, stating the coordinates of the point of intersection. 3

- (b) A body of mass  $m$  kg is travelling in a horizontal straight line so that the resultant force on the body is a resistance force of magnitude  $\frac{1}{10}m\sqrt{1+v}$  when its speed is  $v$  m/s. Initially, the speed of the body is 15 m/s.

- (i) Find the time taken for the body to come to rest. 2  
(ii) Find the distance travelled by the body in coming to rest. 3

- (c) (i) Find the real numbers  $A$  and  $B$  such that

$$\frac{5}{(x+3)(2x+1)} = \frac{A}{x+3} + \frac{B}{2x+1} \quad 2$$

- (ii) Hence, or otherwise, evaluate

$$\int_0^2 \frac{5}{(x+3)(2x+1)} dx \quad 2$$

- (d) Find the points of intersection between the sphere  $S$  and the line  $L$ , given below. 3

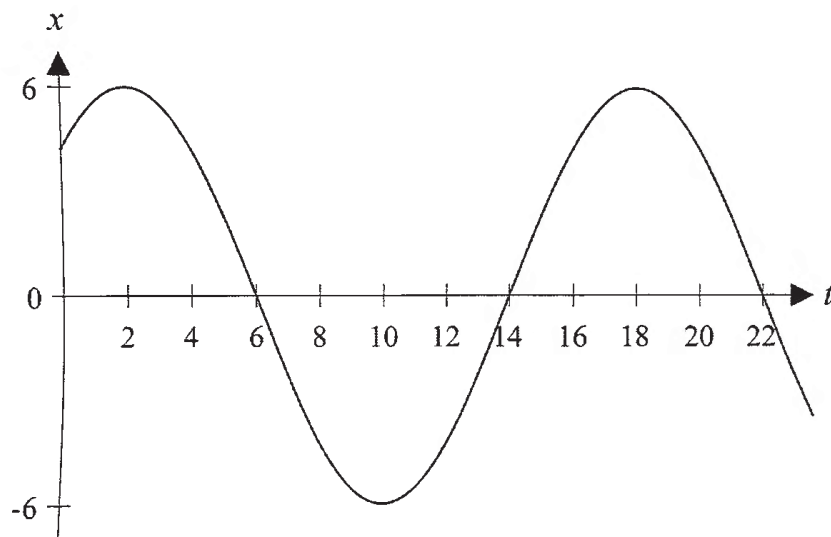
$$S: \left| \mathbf{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$$

$$L: \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet.

- (a) The graph below shows the displacement  $x$  cm from the centre of motion at time  $t$  seconds for a particle performing simple harmonic motion in a straight line.

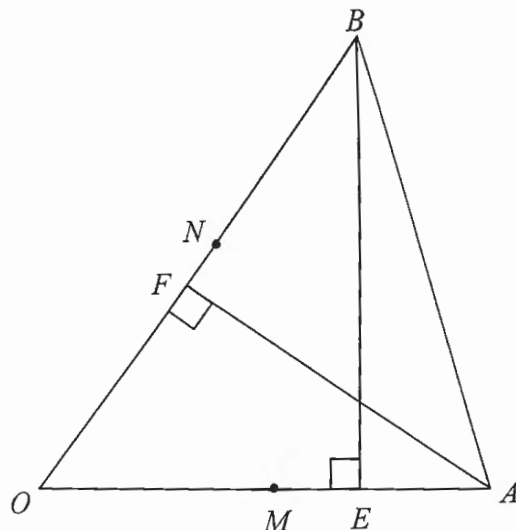


- (i) Express displacement as a function of time in the form  $x = A \sin(nt + \alpha)$  2
- (ii) Find the distance travelled by the particle in the first 30 seconds of its motion after observation began at time  $t = 0$ . 2
- (b) (i) If  $t = \tan \theta$ , prove that  $\tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}$  2
- (ii) If  $\tan \theta \tan 4\theta = 1$ , deduce that  $5t^4 - 10t^2 + 1 = 0$  2
- (iii) Given that  $\theta = \frac{\pi}{10}$  and  $\theta = \frac{3\pi}{10}$  are roots of the equation  $\tan \theta \tan 4\theta = 1$ , find the exact value of  $\tan \frac{\pi}{10}$ . 2
- (c) (i) Show that for  $k \geq 2$ ,  $\tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right) = \tan^{-1}\left(\frac{2}{k^2}\right)$  2
- (ii) Hence, show that  $\lim_{n \rightarrow \infty} \left[ \sum_{k=1}^n \tan^{-1}\left(\frac{2}{k^2}\right) \right] = \frac{3\pi}{4}$  3

**End of Question 14**

**Question 15** (15 marks) Use a SEPARATE writing booklet.

- (a) In  $\triangle OAB$ ,  $BE$  is the altitude from  $B$  to  $OA$  and  $AF$  is the altitude from  $A$  to  $OB$ .  $M$  and  $N$  are the midpoints of  $OA$  and  $OB$  respectively.  $\overrightarrow{OA} = \underline{a}$  and  $\overrightarrow{OB} = \underline{b}$ .



*Not to scale*

Use vector methods to show that  $|OM||OE| = |OM||OF|$

3

- (b) Let  $I_n = \int_0^1 \frac{x^n}{x^2+1} dx$  for  $n \geq 2$

- (i) Show that  $I_n + I_{n-2} = \frac{1}{n-1}$

2

- (ii) Hence, or otherwise, show that  $\int_0^1 \frac{x^2(x-1)^2}{x^2+1} dx = \ln 2 - \frac{2}{3}$

3

- (c) Consider the functions  $f_n(x) = e^x - \left(1 + \sum_{r=1}^n \frac{x^r}{r!}\right)$ ,  $n = 1, 2, 3, \dots$

- (i) Show that  $f_n(0) = 0$  and  $f_{n+1}'(x) = f_n(x)$  for  $n = 1, 2, 3, \dots$

2

- (ii) Show that  $f_1(x) > 0$  for all  $x > 0$  and hence  $1+x < e^x$  for all  $x > 0$

2

- (iii) Use mathematical induction to show that for all positive integers  $n \geq 1$ ,

$$1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} < e^x \text{ for all } x > 0$$

3

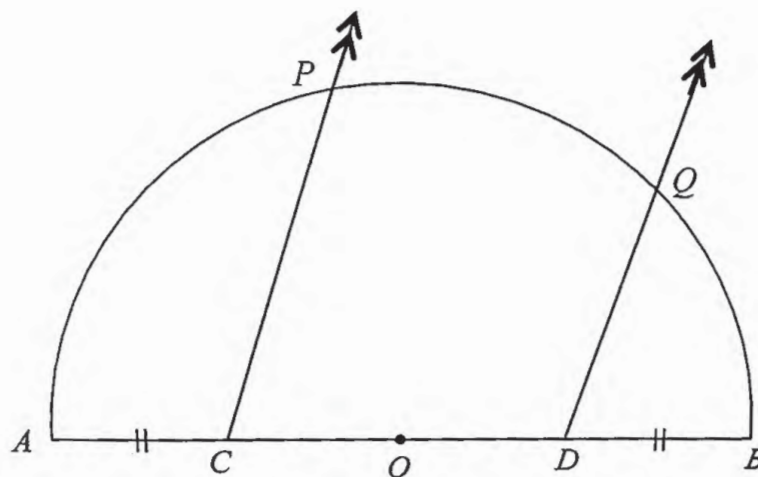
**End of Question 15**

**Question 16** (15 marks) Use a SEPARATE writing booklet.

(a) (i) Show that  $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$ , where  $\cos \theta \neq 0$ ,  $n \in \mathbb{Z}^+$  2

(ii) Hence, show that the roots of  $(1 + z)^2 + (1 - z)^2 = 0$  are  $z = \pm i \tan \frac{\pi}{4}$  when  $z$  is purely imaginary. 2

- (b) A semi-circle is drawn on diameter  $AB$ .  $O$  is the midpoint of  $AB$  and the points  $C$  and  $D$  lie on  $AB$  such that  $AC = BD$ . Parallel lines are drawn through  $C$  and  $D$ , intersecting the semi-circle at  $P$  and  $Q$  respectively.  $\overrightarrow{OC} = \underline{c}$  and  $\overrightarrow{CP} = \underline{p}$ .



(i) Explain why  $\overrightarrow{DQ} = \lambda \underline{p}$  for some scalar  $\lambda > 0$ , then show that  $(1 - \lambda) \underline{p} \cdot \underline{p} + 2 \underline{c} \cdot \underline{p} = 0$  3

(ii) Hence show that  $\angle CPQ = 90^\circ$ . 2

(c) (i) For integer values of  $n$ ,  $I_n = \int_1^e (1 - \ln x)^n dx$  for  $n \geq 0$ . Show that  $I_n = -1 + nI_{n-1}$  for  $n \geq 1$ . 2

(ii) Use mathematical induction to show that  $I_n = n!e - 1 - \sum_{r=1}^n {}^nP_r$  for all integers  $n \geq 1$  4

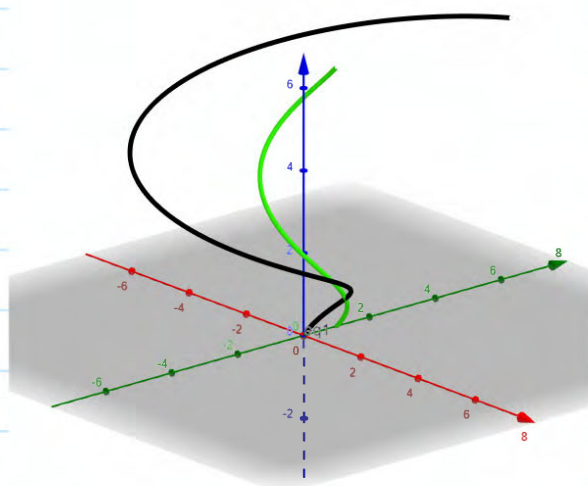
**End of Examination**

## Section 1

- |      |      |      |      |       |
|------|------|------|------|-------|
| 1. A | 2. C | 3. C | 4. C | 5. C  |
| 6. D | 7. D | 8. C | 9. D | 10. A |

## Section 2

### Question 11



(a)  $z = 2 + 2\sqrt{3}i$        $w = 2\sqrt{3} - 2i$

(i)  $|z| = \sqrt{2^2 + (2\sqrt{3})^2} = 4$

$\arg(z) = \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) = \frac{\pi}{3}$

$\therefore z = 4\text{cis}\frac{\pi}{3}$  ✓

$|w| = \sqrt{(2\sqrt{3})^2 + (-2)^2} = 4$

$\arg(w) = \tan^{-1}\left(\frac{-2}{2\sqrt{3}}\right) = -\frac{\pi}{6}$

$\therefore w = 4\text{cis}\left(-\frac{\pi}{6}\right)$  ✓

(ii)  $zw = 4\text{cis}\frac{\pi}{3} \times 4\text{cis}\left(-\frac{\pi}{6}\right)$

$= 16\text{cis}\frac{\pi}{6}$  ✓

$\frac{z}{w} = \frac{4\text{cis}\frac{\pi}{3}}{4\text{cis}\left(-\frac{\pi}{6}\right)}$

$= \text{cis}\frac{\pi}{2}$  ✓

(iii)  $\frac{z}{w} = \text{cis}\frac{\pi}{2} = i \Rightarrow z = iw$  or  $w = -iz$

$\therefore z$  is  $w$  rotated anti-clockwise  $90^\circ$  ✓  
or  $w$  is  $z$  rotated clockwise  $90^\circ$

$$(b) f(z) = z^3 - 11z^2 + 55z - 125 = 0$$

$$(i) \text{ Test } f(5) = 5^3 - 11(5)^2 + 55(5) - 125 = 0$$

$\therefore z=5$  is a root  $\Rightarrow (z-5)$  is a factor ✓

$$\begin{array}{r} z^2 - 6z + 25 \\ z-5 \overline{) z^3 - 11z^2 + 55z - 125} \\ \underline{z^3 - 5z^2} \phantom{+ 55z - 125} \\ -6z^2 + 55z \phantom{- 125} \\ \underline{-6z^2 + 30z} \phantom{- 125} \\ 25z - 125 \\ \underline{25z - 125} \\ 0 \end{array}$$

$$\therefore f(z) = (z-5)(z^2 - 6z + 25) \quad \checkmark$$

$$z = \frac{6 \pm \sqrt{-64}}{2(1)} = \frac{6 \pm 8i}{2} = 3 \pm 4i$$

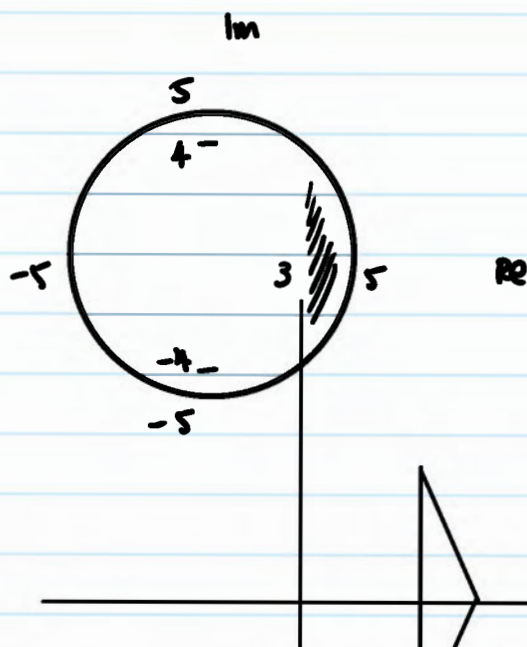
$$\therefore z = 5, 3 \pm 4i \quad \checkmark$$

$$(ii) |5| = 5$$

$$|3+4i| = \sqrt{3^2 + 4^2} = 5$$

$$|3-4i| = \sqrt{3^2 + (-4)^2} = 5$$

$\therefore$  Roots lie on circle  $|z|=5$  ✓



$$\text{Area} = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 2 \times 8$$

$$= 8 \text{ units}^2 \quad \checkmark$$

$$(c) (i) \quad A(3\sin x + 2\cos x) + B(3\cos x - 2\sin x) = 8\sin x + 14\cos x$$

$$3A\sin x + 2A\cos x + 3B\cos x - 2B\sin x = 8\sin x + 14\cos x$$

$$(3A - 2B)\sin x + (2A + 3B)\cos x = 8\sin x + 14\cos x$$

Equating coefficients,

$$3A - 2B = 8 \quad \text{--- (1)}$$

$$2A + 3B = 14 \quad \text{--- (2)}$$

$$(1) \times 2: \quad 6A - 4B = 16 \quad \text{--- (3)}$$

$$(2) \times 3: \quad 6A + 9B = 42 \quad \text{--- (4)}$$

$$(4) - (3): \quad 13B = 26$$

$$B = 2 \quad \text{--- (5)}$$

$$\text{Sub (5) in (1): } 3A - 2(2) = 8$$

$$3A = 12$$

$$A = 4$$

$$\therefore A = 4, B = 2$$

$$(ii) \quad \int_0^{\pi/2} \frac{8\sin x + 14\cos x}{3\sin x + 2\cos x} dx$$

$$= \int_0^{\pi/2} \frac{4(3\sin x + 2\cos x) + 2(3\cos x - 2\sin x)}{3\sin x + 2\cos x} dx$$

$$= \int_0^{\pi/2} 4 + \frac{2(3\sin x + 2\cos x)'}{3\sin x + 2\cos x} dx$$

$$= \left[ 4x + 2\ln |3\sin x + 2\cos x| \right]_0^{\pi/2}$$

$$= \frac{4\pi}{2} + 2\ln |3(1) + 2(0)| - (0 + 2\ln |3(0) + 2(1)|)$$

$$= 2\pi + 2\ln(3/2)$$

# Question 12

(a) GP:  $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$   $a, b, c \in \mathbb{R}$

Common ratio  $r = \frac{\frac{1}{b}}{\frac{1}{a}} = \frac{\frac{1}{c}}{\frac{1}{b}}$

$$\frac{a}{b} = \frac{b}{c}$$

$$\therefore ac = b^2 \text{ --- ①}$$

Now, consider  $(a-b)^2 = a^2 - 2ab + b^2 \geq 0$

$$a^2 + b^2 \geq 2ab \text{ --- ②}$$

Similarly,  $b^2 + c^2 \geq 2bc \text{ --- ③}$

$$a^2 + c^2 \geq 2ac \text{ --- ④}$$

②+③+④:  $2(a^2 + b^2 + c^2) \geq 2(ab + bc + ac)$

$$a^2 + b^2 + c^2 \geq ab + bc + ac \text{ from ①}$$

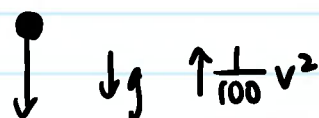
$$\therefore a^2 + c^2 \geq ab + bc, \text{ as required}$$

(b)(i)  $F = ma$

$$g - \frac{1}{100}v^2 = 1 \times \ddot{x}$$

$$\therefore \ddot{x} = g - \frac{1}{100}v^2$$

✓ Show



(ii) When  $\ddot{x} = 0$ ,  $v = V > 0$

$$0 = g - \frac{1}{100}V^2$$

$$\frac{1}{100}V^2 = g$$

$$V^2 = 100g$$

$$\therefore V = 10\sqrt{g}$$

✓ Show

$$(iii) \quad \ddot{x} = v \frac{dv}{dx} = g - \frac{1}{100} v^2$$

$$\frac{dv}{dx} = \frac{100g - v^2}{100v}$$

$$\frac{dx}{dv} = \frac{100v}{100g - v^2}$$

$$x = \int \frac{100v}{100g - v^2} dv$$

$$= -50 \ln |100g - v^2| + C$$

$$\text{At } t=0, x=0, v=0$$

$$0 = -50 \ln |100g - 0^2| + C$$

$$C = 50 \ln 100g$$

$$\therefore x = 50 \ln \left| \frac{100g}{100g - v^2} \right|$$

$$\frac{x}{50} = \ln \left| \frac{100g}{100g - v^2} \right|$$

$$e^{\frac{x}{50}} = \frac{100g}{100g - v^2}$$

$$100g - v^2 = 100g e^{-x/50}$$

$$v^2 = 100g (1 - e^{-x/50})$$

$$\therefore v^2 = V^2 (1 - e^{-x/50})$$

$$(iv) \quad \text{Sub } v = V/2$$

$$\frac{V^2}{4} = V^2 (1 - e^{-x/50})$$

$$e^{-x/50} = 3/4$$

$$-\frac{x}{50} = \ln \frac{3}{4}$$

$$x = 50 \ln \frac{4}{3}$$

$$\therefore x = 14.38 \text{ m (2dp)}$$

$$(c) \int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx$$

$$= \int_a^{\frac{1}{a}} \frac{\ln(\frac{1}{u})}{1+(\frac{1}{u})^2} \cdot \frac{-1}{u^2} du$$

$$= \int_{\frac{1}{a}}^a \frac{-\ln u}{\frac{u^2+1}{u^2}} \cdot \frac{1}{u^2} du \quad \checkmark$$

$$= \int_{\frac{1}{a}}^a \frac{-\ln u}{1+u^2} du$$

$$= \int_{\frac{1}{a}}^a \frac{-\ln x}{1+x^2} dx \quad \text{Dummy variable}$$

Let  $u = \frac{1}{x} \rightarrow x = \frac{1}{u}$   
 $du = -\frac{1}{x^2} dx$   
 $dx = -x^2 du$   
 $dx = -\frac{1}{u^2} du$   
 $\begin{cases} x = \frac{1}{a}, & u = a \\ x = a, & u = \frac{1}{a} \end{cases}$

$$2 \int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx = 0 \quad \checkmark$$

$$\therefore \int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx = 0$$

$$(ii) \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\tan^{-1} x}{x} dx$$

$u = \tan^{-1} x \quad v' = \frac{1}{x}$   
 $u' = \frac{1}{1+x^2} \quad v = \ln x$

$$= \left[ \ln x \cdot \tan^{-1} x \right]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} - \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx \quad \checkmark$$

$$= \ln \sqrt{3} \cdot \frac{\pi}{3} - \ln\left(\frac{1}{\sqrt{3}}\right) \cdot \frac{\pi}{6} - \underline{0} \quad \text{from (i)} \quad \checkmark$$

$$= \frac{\pi \ln \sqrt{3}}{3} + \frac{\pi \ln \sqrt{3}}{6}$$

$$= \frac{\pi \ln \sqrt{3}}{2}$$

$$= \frac{\pi \ln 3}{4} \quad \checkmark$$

Question 13

$$(a) L_1: \underline{r} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad L_2: \underline{r} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 2 \times 1 + -1 \times 1 + 1 \times -1 = 0$$

$$\therefore L_1 \perp L_2$$

$$\text{Then, POI when } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$3 + 2\lambda = -1 + \mu \quad \text{--- (1)}$$

$$2 - \lambda = 1 + \mu \quad \text{--- (2)}$$

$$-1 + \lambda = -\mu \quad \text{--- (3)}$$

$$\textcircled{1} - \textcircled{2}: \quad 1 + 3\lambda = -2$$

$$\lambda = -1 \quad \text{--- (4)}$$

$$\text{Sub (4) in (2): } 2 - (-1) = 1 + \mu$$

$$\mu = 2$$

$$\text{Test } \lambda = -1, \mu = 2 \text{ in (3): } \text{LHS} = -1 - 1 = -2$$

$$\text{RHS} = -2$$

$$\therefore \text{POI at } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} - 1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \\ = \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$$

$$(b)(i) \quad F = ma$$

$$\frac{-1}{10} \sqrt{1+v} = m \ddot{x}$$

$$\ddot{x} = \frac{dv}{dx} = \frac{-1}{10} \sqrt{1+v}$$

$$\frac{dt}{dv} = \frac{-10}{\sqrt{1+v}} \quad \checkmark$$

$$\int_0^T dt = \int_{15}^0 -10(1+v)^{-1/2} dv$$

$$[t]_0^T = 10 \left[ \frac{(1+v)^{1/2}}{1/2} \right]_0^{15}$$

$$T - 0 = 20(\sqrt{16} - \sqrt{1})$$

$$\therefore T = 60 \text{ seconds} \quad \checkmark$$

$$(ii) \quad \ddot{x} = v \frac{dv}{dx} = \frac{-1}{10} \sqrt{1+v}$$

$$\frac{dv}{dx} = \frac{-\sqrt{1+v}}{10v}$$

$$\frac{dx}{dv} = \frac{-10v}{\sqrt{1+v}} \quad \checkmark$$

$$\int_0^D dx = \int_{15}^0 \frac{-10v}{\sqrt{1+v}} dv$$

$$[x]_0^D = 10 \int_0^{15} \frac{1+v-1}{\sqrt{1+v}} dv$$

$$D - 0 = 10 \int_0^{15} (1+v)^{1/2} - (1+v)^{-1/2} dv \quad \checkmark$$

$$D = \left[ \frac{(1+v)^{3/2}}{3/2} - \frac{(1+v)^{1/2}}{1/2} \right]_0^{15}$$

$$= \frac{2}{3} (16\sqrt{16} - \sqrt{1}) - 2(\sqrt{16} - \sqrt{1})$$

$$\therefore D = 360 \text{ m} \quad \checkmark$$

$$(c)(i) \quad \frac{5}{(x+3)(2x+1)} = \frac{A}{x+3} + \frac{B}{2x+1}$$

$$5 = A(2x+1) + B(x+3)$$

$$\text{Sub } x = -\frac{1}{2}$$

$$5 = A(0) + B\left(\frac{5}{2}\right)$$

$$B = 2$$

$$\text{Sub } x = -3$$

$$5 = A(-5) + B(0)$$

$$A = -1$$

$$\therefore A = -1, B = 2$$

$$(ii) \quad \int_0^2 \frac{5}{(x+3)(2x+1)} dx = \int_0^2 \frac{-1}{x+3} + \frac{2}{2x+1} dx$$

$$= \left[ -\ln|x+3| + \ln|2x+1| \right]_0^2$$

$$= \left[ \ln \left| \frac{2x+1}{x+3} \right| \right]_0^2$$

$$= \ln\left(\frac{5}{3}\right) - \ln\left(\frac{1}{3}\right)$$

$$= 0 - \ln\left(\frac{1}{3}\right)$$

$$= -\ln\left(\frac{1}{3}\right)$$

$$= \ln 3$$

$$(d) \quad S: \left| \vec{r} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3 \quad \text{and} \quad L: \vec{r} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

POI when

$$\left| \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \right| = 3$$

$$\left| \begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} \right| = 3$$

$$|\underline{a}|^2 = \underline{a} \cdot \underline{a}$$



$$\begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} \cdot \begin{pmatrix} 1+\lambda \\ 1+\lambda \\ -1+2\lambda \end{pmatrix} = 9$$

$$(1+\lambda)^2 + (1+\lambda)^2 + (-1+2\lambda)^2 = 9$$

$$1+2\lambda+\lambda^2 + 1+2\lambda+\lambda^2 + 1-4\lambda+4\lambda^2 = 9$$

$$6\lambda^2 = 6$$

$$\lambda = \pm 1$$



Sub  $\lambda = 1$  in  $L$ :

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Sub  $\lambda = -1$  in  $L$ :

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$$

✓ Both POIs

# Question 14

(a) (i)  $x = A \sin(\omega t + \alpha)$

$$A = 6$$

$$T = \frac{2\pi}{\omega} = 16$$

$$\omega = \frac{\pi}{8} \quad \checkmark$$

$$\text{At } t=2, \quad x=6$$

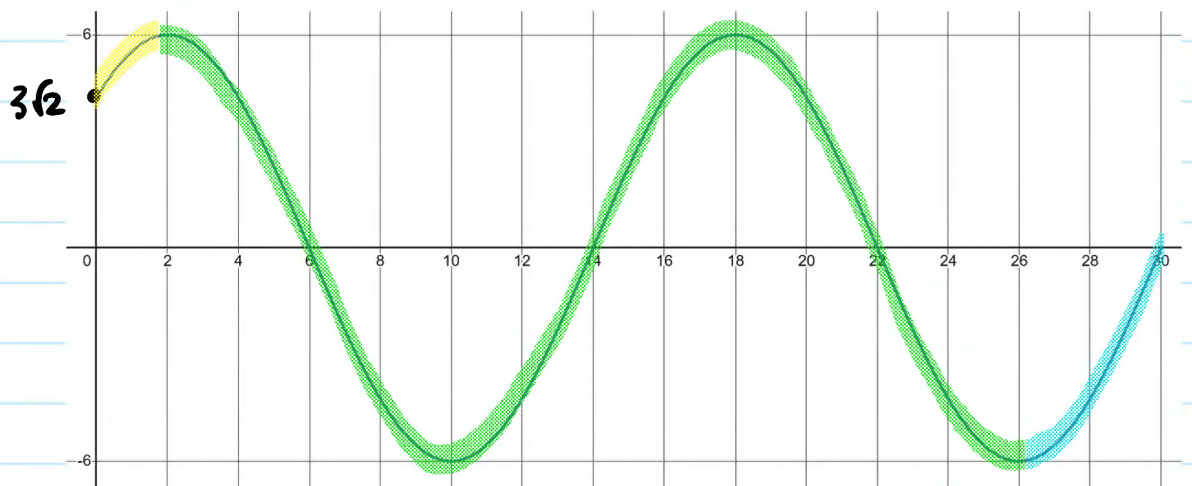
$$6 = 6 \sin\left(\frac{\pi}{4} + \alpha\right)$$

$$\frac{\pi}{4} + \alpha = \sin^{-1}(1) = \frac{\pi}{2}$$

$$\alpha = \frac{\pi}{4}$$

$$\therefore x = 6 \sin\left(\frac{\pi t}{8} + \frac{\pi}{4}\right) \quad \checkmark$$

(ii)



$$\text{At } t=0, \quad x = 6 \sin\left(0 + \frac{\pi}{4}\right) = \frac{6}{\sqrt{2}} = 3\sqrt{2} \quad \checkmark$$

$$\text{Distance} = (6 - 3\sqrt{2}) + 3 \times 12 + 6$$

$$= 48 - 3\sqrt{2} \text{ metres} \quad \checkmark$$

$$(b) (i) \text{ RHP: } \tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}$$

$$\text{LHS} = \tan 4\theta$$

$$= \tan(2 \times 2\theta)$$

$$= \frac{2 \tan 2\theta}{1 - \tan^2 2\theta}$$

$$= \frac{2 \left( \frac{2 \tan \theta}{1 - \tan^2 \theta} \right)}{1 - \left( \frac{2 \tan \theta}{1 - \tan^2 \theta} \right)^2}$$

Sub  $t = \tan \theta$



$$= \frac{\frac{4t}{1-t^2}}{1 - \frac{4t^2}{(1-t^2)^2}}$$

$$= \frac{\frac{4t}{1-t^2}}{\frac{1-4t^2}{(1-t^2)^2}}$$

$$= \frac{4t(1-t^2)}{1-6t^2+t^4}$$



$$= \text{RHS}$$

$$(ii) \tan \theta \tan 4\theta = 1$$

$$t \times \frac{4t(1-t^2)}{1-6t^2+t^4} = 1$$



$$4t^2(1-t^2) = 1-6t^2+t^4$$

$$4t^2 - 4t^4 = 1-6t^2+t^4$$



$$\therefore 5t^4 - 10t^2 + 1 = 0$$

$$(iii) t^2 = \frac{10 \pm \sqrt{80}}{2(5)} = \frac{5 \pm 2\sqrt{5}}{5}$$

$$t = \pm \sqrt{\frac{5 \pm 2\sqrt{5}}{5}}$$



$$\text{But } 0 < \tan \frac{\pi}{10} < \tan \frac{3\pi}{10}$$

$$\therefore \tan \frac{\pi}{10} = \sqrt{\frac{5-2\sqrt{5}}{5}}$$



(c)(i) IP:  $\underbrace{\tan^{-1}\left(\frac{1}{k-1}\right)}_{\alpha} - \underbrace{\tan^{-1}\left(\frac{1}{k+1}\right)}_{\beta} = \tan^{-1}\left(\frac{2}{k^2}\right)$  for  $k \geq 2$

Consider  $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$   
 $= \frac{\frac{1}{k-1} - \frac{1}{k+1}}{1 + \left(\frac{1}{k-1}\right)\left(\frac{1}{k+1}\right)}$

$\alpha = \tan^{-1}\left(\frac{1}{k-1}\right)$   
 $\tan \alpha = \frac{1}{k-1}$   
 $\beta = \tan^{-1}\left(\frac{1}{k+1}\right)$   
 $\tan \beta = \frac{1}{k+1}$

$= \frac{\frac{k+1 - k+1}{(k-1)(k+1)}}{\frac{k^2 - 1 + 1}{(k-1)(k+1)}}$   
 $= \frac{2}{k^2}$

$\alpha - \beta = \tan^{-1}\left(\frac{2}{k^2}\right)$

$\therefore \tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right) = \tan^{-1}\left(\frac{2}{k^2}\right)$

(ii)  $\sum_{k=1}^{\infty} \tan^{-1}\left(\frac{2}{k^2}\right)$

$= \tan^{-1} 2 + \sum_{k=2}^{\infty} \tan^{-1}\left(\frac{1}{k-1}\right) - \tan^{-1}\left(\frac{1}{k+1}\right)$

$= \tan^{-1} 2 + \underbrace{\tan^{-1}(1) - \tan^{-1}\left(\frac{1}{2}\right)}_{k=2} + \underbrace{\tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{4}\right)}_{k=3} + \underbrace{\tan^{-1}\left(\frac{1}{4}\right) - \tan^{-1}\left(\frac{1}{6}\right)}_{k=4}$   
 $+ \underbrace{\tan^{-1}\left(\frac{1}{6}\right) - \tan^{-1}\left(\frac{1}{8}\right)}_{k=5} + \dots + \underbrace{\tan^{-1}\left(\frac{1}{n-2}\right) - \tan^{-1}\left(\frac{1}{n}\right)}_{k=n-1} + \underbrace{\tan^{-1}\left(\frac{1}{n-1}\right) - \tan^{-1}\left(\frac{1}{n+1}\right)}_{k=n}$

$= \tan^{-1} 2 + \frac{\pi}{4} + \tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{n}\right) - \tan^{-1}\left(\frac{1}{n+1}\right)$

$\lim_{n \rightarrow \infty} \left[ \sum_{k=1}^n \tan^{-1}\left(\frac{2}{k^2}\right) \right] = \lim_{n \rightarrow \infty} \left( \frac{\pi}{4} + \tan^{-1} 2 + \tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{n}\right) - \tan^{-1}\left(\frac{1}{n+1}\right) \right)$

$= \frac{\pi}{4} + \tan^{-1} 2 + \tan^{-1}\left(\frac{1}{2}\right) - \lim_{n \rightarrow \infty} \left( \tan^{-1}\left(\frac{1}{n}\right) + \tan^{-1}\left(\frac{1}{n+1}\right) \right)$

$= \frac{\pi}{4} + \frac{\pi}{2} - 0$   $\tan(A+B) = \frac{2 + \frac{1}{2}}{1 - (2 \times \frac{1}{2})} = \text{undefined}$

$= \frac{3\pi}{4}$

OR...  $\tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}, x > 0$

Question 15

$$(a) \vec{OM} = \frac{1}{2} \vec{OA} = \frac{1}{2} \underline{a}$$

$$\vec{OE} = \lambda \vec{OA} = \lambda \underline{a} \quad \lambda \text{ scalar}$$

$$\vec{ON} = \frac{1}{2} \vec{OB} = \frac{1}{2} \underline{b}$$

$$\vec{OF} = \mu \vec{OB} = \mu \underline{b} \quad \mu \text{ scalar}$$

$$\text{Now, } |\vec{OM}| |\vec{OE}| = \frac{\lambda}{2} |\underline{a}|^2$$

$$|\vec{ON}| |\vec{OF}| = \frac{\mu}{2} |\underline{b}|^2$$

$$\text{Also, } AF \perp OB, \text{ so } \vec{AF} \cdot \vec{OB} = 0$$

$$(\mu \underline{b} - \underline{a}) \cdot \underline{b} = 0$$

$$\mu \underline{b} \cdot \underline{b} - \underline{a} \cdot \underline{b} = 0$$

$$\underline{a} \cdot \underline{b} = \mu \underline{b} \cdot \underline{b} = \mu |\underline{b}|^2$$

$$\text{Similarly, } BE \perp OA, \text{ so } \vec{BE} \cdot \vec{OA} = 0$$

$$(\underline{b} - \lambda \underline{a}) \cdot \underline{a} = 0$$

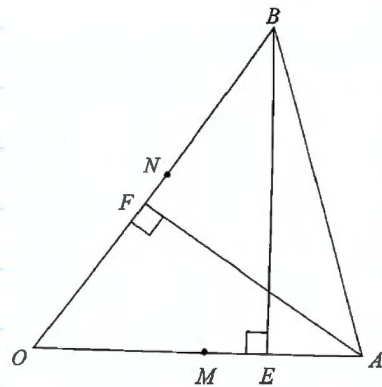
$$\underline{a} \cdot \underline{b} - \lambda \underline{a} \cdot \underline{a} = 0$$

$$\underline{a} \cdot \underline{b} = \lambda \underline{a} \cdot \underline{a} = \lambda |\underline{a}|^2$$

$$\text{Then } \underline{a} \cdot \underline{b} = \lambda |\underline{a}|^2 = \mu |\underline{b}|^2$$

$$\frac{\lambda}{2} |\underline{a}|^2 = \frac{\mu}{2} |\underline{b}|^2$$

$$\therefore |\vec{OM}| |\vec{OE}| = |\vec{ON}| |\vec{OF}|$$



$$(b) \quad I_n = \int_0^1 \frac{x^n}{x^2+1} dx$$

$$(i) \quad LHS = I_n + I_{n-2}$$

$$= \int_0^1 \frac{x^n}{x^2+1} dx + \int_0^1 \frac{x^{n-2}}{x^2+1} dx$$

$$= \int_0^1 \frac{x^{n-2} (x^2+1)}{x^2+1} dx$$

$$= \left[ \frac{x^{n-1}}{n-1} \right]_0^1$$

$$= \frac{1}{n-1} - 0$$

$$= \frac{1}{n-1}$$

$$= RHS$$

$$(ii) \quad \int_0^1 \frac{x^2(x-1)^2}{x^2+1} dx = \int_0^1 \frac{x^2(x^2-2x+1)}{x^2+1} dx$$

$$= \int_0^1 \frac{x^4}{x^2+1} - 2 \frac{x^3}{x^2+1} + \frac{x^2}{x^2+1} dx$$

$$= I_4 - 2I_3 + I_2$$

$$= \frac{1}{3} - 2\left(\frac{1}{2} - I_1\right) \quad \text{from (i)}$$

$$= \frac{1}{3} - 1 + \int_0^1 \frac{2x}{x^2+1} dx$$

$$= -\frac{2}{3} + \left[ \ln(x^2+1) \right]_0^1$$

$$= -\frac{2}{3} + \ln 2 - \ln 1$$

$$= \ln 2 - \frac{2}{3}$$

$$(i) f_n(x) = e^x - \left(1 + \sum_{r=1}^n \frac{x^r}{r!}\right) \quad n=1, 2, 3, \dots$$

$$(ii) f_n(0) = e^0 - \left(1 + \sum_{r=1}^n \frac{0^r}{r!}\right)$$

$$= 1 - 1 - 0$$

$$= 0$$

$$f_{n+1}(x) = e^x - \left(1 + \sum_{r=1}^{n+1} \frac{x^r}{r!}\right)$$

$$= e^x - 1 - \left(\frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!}\right)$$

$$f_{n+1}'(x) = e^x - 0 - \left(\frac{1}{1!} + \frac{2x}{2!} + \frac{3x^2}{3!} + \frac{4x^3}{4!} + \dots + \frac{nx^{n-1}}{n!} + \frac{(n+1)x^n}{(n+1)!}\right)$$

$$= e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{n-1}}{(n-1)!} + \frac{x^n}{n!}\right)$$

$$= e^x - \left(1 + \sum_{r=1}^n \frac{x^r}{r!}\right)$$

$$= f_n(x)$$

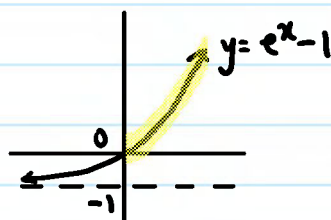
$$(ii) f_1(x) = e^x - \left(1 + \frac{x^1}{1!}\right)$$

$$= e^x - 1 - x$$

$$f_1'(x) = e^x - 0 - 1$$

$$= e^x - 1$$

$$> 0 \text{ for } x > 0$$



$\therefore f_1(x)$  is increasing for  $x > 0$

Since  $f_1(0) = 0$ , then  $f_1(x) > 0$  for all  $x > 0$

$$e^x - 1 - x > 0$$

$$\therefore 1 + x < e^x \text{ for all } x > 0$$

(iii) RTP:  $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} < e^x$  for  $x > 0$ .

Step 1 For  $n=1$

$$\text{LHS} = 1+x$$

$$\text{RHS} = e$$

$$[1+x < e^x \text{ from (ii)}]$$

LHS < RHS, statement is true for  $n=1$  ✓

Step 2 Assume statement is true for  $n=k$

$$\text{i.e. } 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!} < e^x$$

$$\Rightarrow e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!}\right) = f_k(x) > 0 \quad \checkmark$$

Step 3 Attempt to prove statement is true for  $n=k+1$

$$\text{i.e. } 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{k+1}}{(k+1)!} < e^x$$

$$\text{RHS} - \text{LHS} = e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{k+1}}{(k+1)!}\right)$$

$$= f_{k+1}(x)$$

$$> 0, \text{ since } f_{k+1}(0) = 0$$

$$\text{and } f_{k+1}'(x) = f_k(x) > 0 \quad \text{(i) and from assumption} \quad \checkmark$$

$$\therefore \text{RHS} > \text{LHS}$$

$\therefore$  Statement is true for  $n=k+1$  if it is true for  $n=k$

Step 4 By the principle of mathematical induction, the statement is true for integers  $n \geq 1$ .

Question 16

(a) (i) RP:  $(1+i\tan\theta)^n + (1-i\tan\theta)^n = \frac{2\cos n\theta}{\cos^n\theta}$   $\cos\theta \neq 0, n \in \mathbb{Z}^+$

$$\text{LHS} = (1+i\tan\theta)^n + (1-i\tan\theta)^n$$

$$= \left(1 + \frac{i\sin\theta}{\cos\theta}\right)^n + \left(1 - \frac{i\sin\theta}{\cos\theta}\right)^n$$

$$= \left(\frac{\cos\theta + i\sin\theta}{\cos\theta}\right)^n + \left(\frac{\cos\theta - i\sin\theta}{\cos\theta}\right)^n$$

$$= \left(\frac{\text{cis}\theta}{\cos\theta}\right)^n + \left(\frac{\text{cis}(-\theta)}{\cos\theta}\right)^n$$

$$= \frac{\text{cis}(n\theta) + \text{cis}(-n\theta)}{\cos^n\theta}$$

$$= \frac{\cos(n\theta) + i\sin(n\theta) + \cos(n\theta) - i\sin(n\theta)}{\cos^n\theta}$$

$$\begin{aligned}\cos(-n\theta) &= \cos(n\theta) \\ \sin(-n\theta) &= -\sin(n\theta)\end{aligned}$$

$$= \frac{2\cos(n\theta)}{\cos^n\theta}$$

$$= \text{RHS}$$

(ii) Let  $z = i\tan\theta, n=2$

$$(1+z)^2 + (1-z)^2 = 0 \rightarrow \text{Quadratic eqn, 2 roots}$$

$$\frac{2\cos 2\theta}{\cos^2\theta} = 0$$

$$\cos 2\theta = 0$$

$$2\theta = \pi/2, 3\pi/2$$

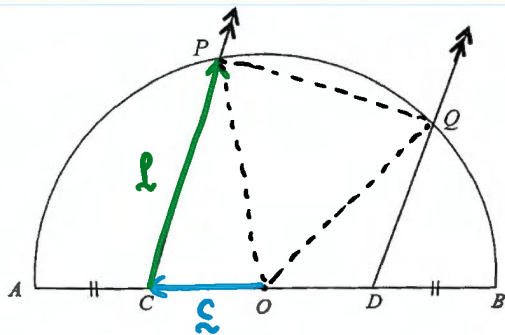
$$\theta = \pi/4, 3\pi/4$$

$$z_1 = i\tan\frac{\pi}{4}$$

$$z_2 = i\tan\frac{3\pi}{4} = i\tan\left(-\frac{\pi}{4}\right) = -i\tan\frac{\pi}{4}$$

$$\therefore z = \pm i\tan\frac{\pi}{4}$$

(b)



$$(i) \quad \vec{DQ} = \lambda \vec{CP} \quad (\text{DQ} \parallel \text{CP})$$

$$\therefore \vec{DQ} = \lambda \vec{r} \quad \text{for } \lambda > 0 \text{ scalar}$$

✓ Explain

$$\begin{aligned} \text{Now, } \vec{OP} &= \vec{OC} + \vec{CP} \\ &= \vec{s} + \vec{r} \\ \vec{OQ} &= \vec{OD} + \vec{DQ} \\ &= -\vec{s} + \lambda \vec{r} \end{aligned}$$

Then,  $|\vec{OP}| = |\vec{OQ}|$  (equal radii of circle)

$$(\vec{s} + \vec{r}) \cdot (\vec{s} + \vec{r}) = (-\vec{s} + \lambda \vec{r}) \cdot (-\vec{s} + \lambda \vec{r})$$

$$\cancel{s} \cdot \cancel{s} + 2\vec{s} \cdot \vec{r} + \vec{r} \cdot \vec{r} = \cancel{s} \cdot \cancel{s} - 2\lambda \vec{s} \cdot \vec{r} + \lambda^2 \vec{r} \cdot \vec{r}$$

$$\vec{r} \cdot \vec{r} (1 - \lambda^2) = -2\vec{s} \cdot \vec{r} (1 + \lambda)$$

$$\vec{r} \cdot \vec{r} (1 - \lambda)(1 + \lambda) = -2\vec{s} \cdot \vec{r} (1 + \lambda)$$

$$\therefore (1 - \lambda) \vec{r} \cdot \vec{r} + 2\vec{s} \cdot \vec{r} = 0$$

$$(ii) \quad \vec{PQ} = \vec{PC} + \vec{CB} + \vec{DQ}$$

$$= -\vec{r} - 2\vec{s} + \lambda \vec{r}$$

$$= (\lambda - 1) \vec{r} - 2\vec{s}$$

$$\text{Now, } \vec{CP} \cdot \vec{PQ} = \vec{r} \cdot ((\lambda - 1) \vec{r} - 2\vec{s})$$

$$= (\lambda - 1) \vec{r} \cdot \vec{r} - 2\vec{s} \cdot \vec{r}$$

$$= -[(1 - \lambda) \vec{r} \cdot \vec{r} + 2\vec{s} \cdot \vec{r}]$$

$$= 0$$

from (i)

$$\therefore \text{CP} \perp \text{PQ}$$

$$\therefore \angle CPQ = 90^\circ$$

$$(c)(i) \quad I_n = \int_1^e (1 - \ln x)^n dx$$

$$u = (1 - \ln x)^n \quad v' = 1$$

$$u' = \frac{n(1 - \ln x)^{n-1}}{-x} \quad v = x$$

$$= \left[ x(1 - \ln x)^n \right]_1^e - \int_1^e \frac{n(1 - \ln x)^{n-1}}{-x} \cdot x dx$$

$$= e(1 - \ln e)^n - 1(1 - \ln 1)^n + n \int_1^e (1 - \ln x)^{n-1} dx$$

$$= 0 - 1 + n I_{n-1}$$

$$= -1 + n I_{n-1}$$

$$(ii) \text{ RTP: } I_n = n!e - 1 - \sum_{r=1}^n n P_r \quad n \geq 1$$

Step 1 for  $n=1$

$$\text{LHS} = I_1$$

$$= -1 + I_0$$

$$= -1 + \int_1^e dx$$

$$= -1 + e - 1$$

$$= e - 2$$

$$\text{RHS} = 1!e - 1 - {}^1P_1$$

$$= e - 1 - 1$$

$$= e - 2$$

$\therefore \text{LHS} = \text{RHS}$ , Statement is true for  $n=1$

Step 2 Assume statement is true for  $n=k$

$$\text{i.e. } I_k = k!e - 1 - \sum_{r=1}^k {}^kP_r$$

Step 3 Attempt to prove true for  $n=k+1$

$$\text{i.e. } I_{k+1} = (k+1)!e - 1 - \sum_{r=1}^{k+1} {}^{k+1}P_r$$

$$\text{LHS} = I_{k+1}$$

$$= -1 + (k+1)I_k$$

$$= -1 + (k+1) \left[ k!e - 1 - \sum_{r=1}^k {}^kP_r \right]$$

$$= -1 + (k+1)!e - (k+1) - (k+1) [{}^kP_1 + {}^kP_2 + {}^kP_3 + \dots + {}^kP_k]$$

PTD  $\rightarrow$

$$= -1 + (k+1)!e - (k+1) - (k+1) \left[ \frac{k!}{(k-1)!} + \frac{k!}{(k-2)!} + \frac{k!}{(k-3)!} + \dots + \frac{k!}{(k-k)!} \right]$$

$$= -1 + (k+1)!e - (k+1) - \left( \frac{(k+1)!}{(k+1-2)!} + \frac{(k+1)!}{(k+1-3)!} + \frac{(k+1)!}{(k+1-4)!} + \dots + \frac{(k+1)!}{(k+1-(k+1))!} \right)$$

$$= -1 + (k+1)!e - (k+1) - \sum_{r=2}^{k+1} (k+1)p_r$$

$$= -1 + (k+1)!e - (k+1) - \left( \sum_{r=1}^{k+1} (k+1)p_r - (k+1)p_1 \right)$$

$$= -1 + (k+1)!e - \cancel{(k+1)} - \sum_{r=1}^{k+1} (k+1)p_r + \cancel{(k+1)}$$

$$= (k+1)!e - 1 - \sum_{r=1}^{k+1} (k+1)p_r$$

$$= \text{RHS}$$

$\therefore$  Statement is true for  $n=k+1$  if it is true for  $n=k$

Step 4 By the principle of mathematical induction, statement is true for integers  $n \geq 1$ .